This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

1. Our design requirements are that the pill box has to be; able to send notifications/display a message to the user on when to take their meds; big enough to hold all the mechanics and pills, while not taking up too much space; have enough space for all the medication. The box must notify the user because that is its main purpose, to remind the user to take their medication. We feel that a notification is the best way to do this. We want the box to be a single unit and have the mechanics attached for space efficiency. Holding all of the user's pills is extremely important.
2. (and 3. And 4.) Our design requirements are:
    1. The box must be big enough to fit all of the user's medication. (high)
        a. The pills need to be in a singular place.
        b. This will be executed with a plexiglass case. Measuring about 4 inches by 7 inches with a depth of 3 inches. This is a high priority because usually patients have to take multiple forms of medication and it is important that there is enough space for each day.
    2. The box must notify the user via displayed message. (high)
        a. For those who struggle with remembering things, or have to take meds multiple times during the day this is vital.
        b. This would be executed with an Ardiuno display at the front of the pill box. This will coded to display the message "it's time ot take your medication", in order to remind the patient. This is a high priority because it is one of the main mechanical functions of the pill box.
    3. The mechanics must be able to fit onto the box to optimize space and a sensor to check when the box has been opened (high)
        a. In order for this to work at its best it must be able to stay in one place without multiple pieces of mechanics around.
        b. This would be done with an Arduino sensor and would be the other main part of mechanics. And would be what triggers the message of when the patient has to take thier medication.
- The box being able to hold the user's medication is the main function of a pill box but the difference is that it must notify the individual in order to remind them to take their meds.
- Our stakeholders are the main reasons we have these requirements and theoretically they would agree with and value these requirements.

5. These design requirements would lead to a solution that would be built and achieve the needs of our stakeholders because it takes into account their main needs when it comes to being reminded to take their medication. It is mechanically not super complex

so it would be easier to repair than other machines.
6. When we presented our project, one of our evaluators expressed how this would have been very helpful when he was taking care of his father in law. He knew the struggles of having to remind someone to take their medication. How often his father in law would forget and thus make his medication less effective. We would need to add to the list, prioritizing not only ample space in each section but areas for medication that must been taken at different times: morning, afternoon and night.